BSc in Artificial Intelligence

A.A. 2025/-26

Exam Project 2:

Customer Churn Prediction

Made by
Lucrezia Bernini

DATA OVERVIEW

Churn Analysis

5 621
Total Customers

0.95
Best F1-Score (XGB)

16.8%
Churn Rate

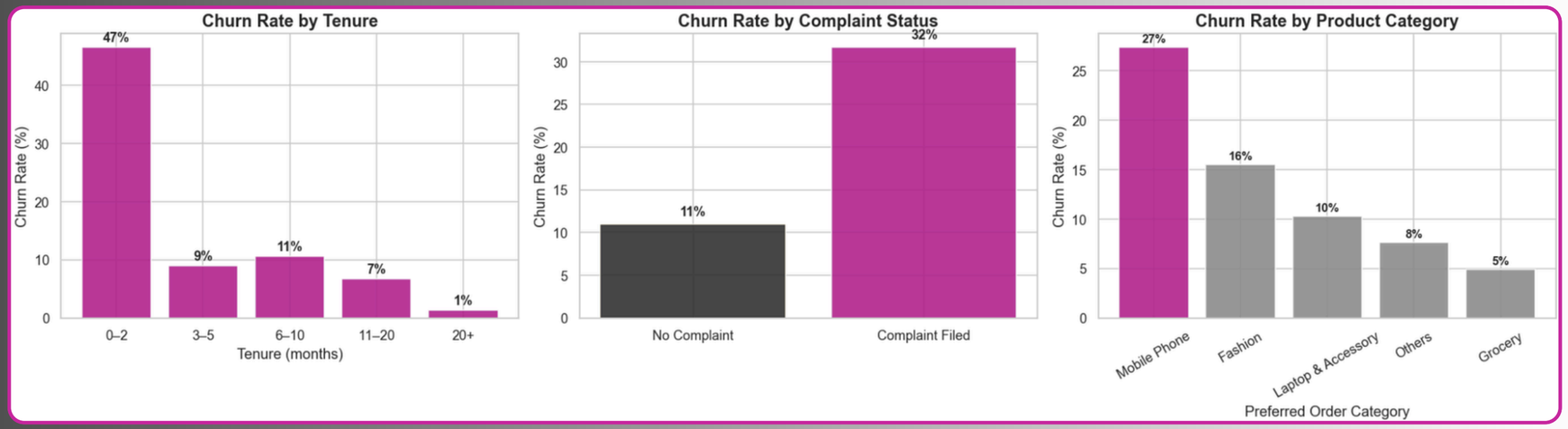
5.92×
Lift at Top 10%

Top 3 Risk Drivers

1. **Tenure —** The #1 predictor across all models. High Risk customers have an avg tenure of just 3.9 months — the company loses customers during onboarding.

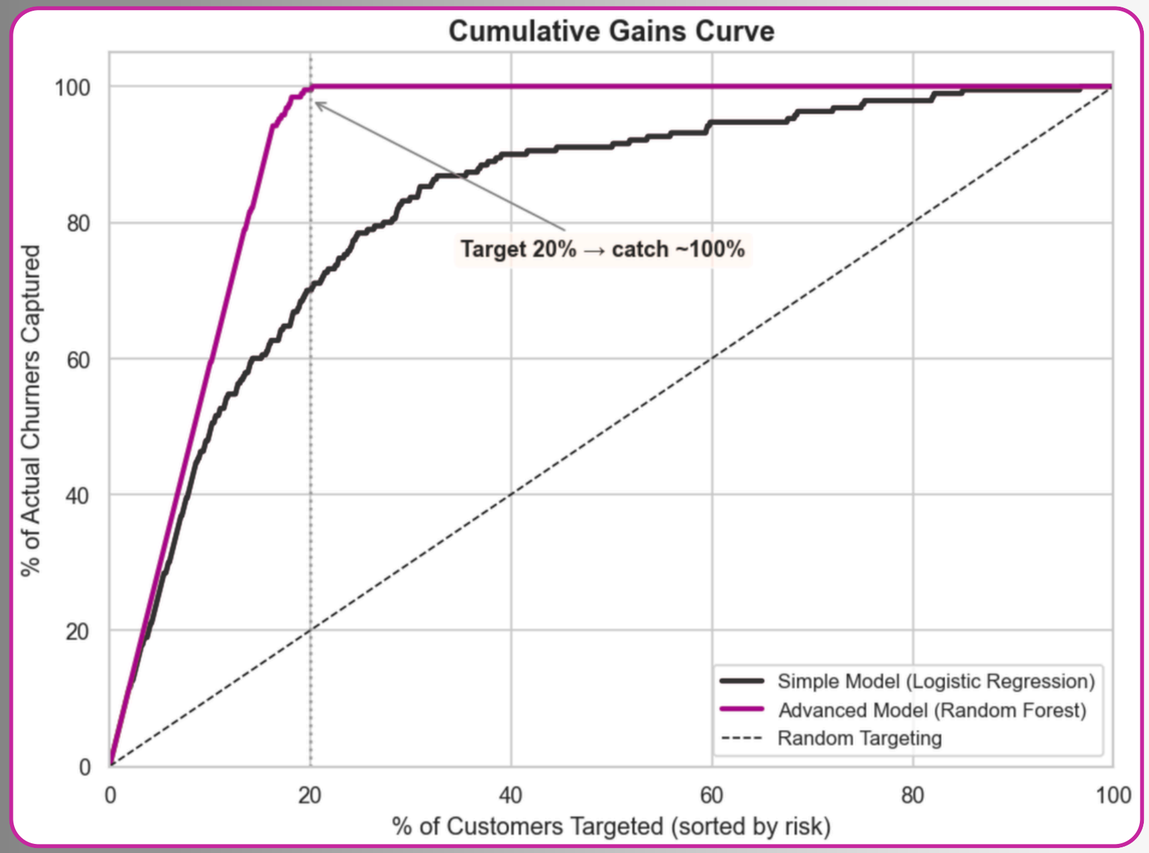
2. **Complain —** 54% of High Risk customers filed a complaint, vs only 23% of Low Risk. An unresolved complaint is a direct trigger for churn.

3. **Product Category —** Customers who primarily order Mobile Phone show the highest churn rate (27%). These are typically high-frequency, low-commitment buyers who are harder to retain and need dedicated re-engagement strategies.



How Well Can We Predict Churn?

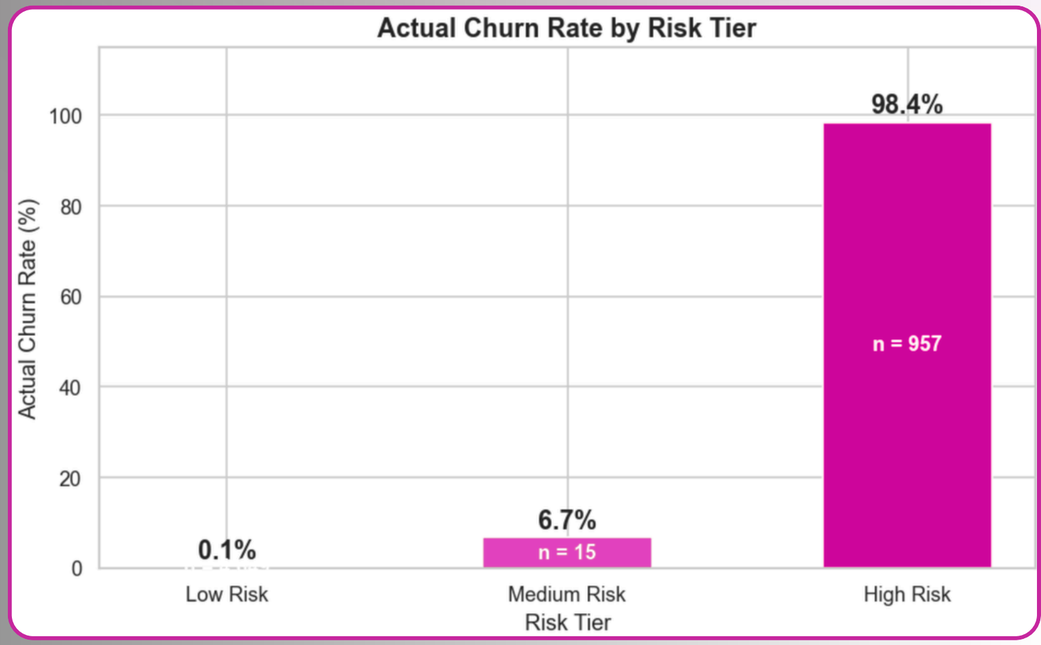
	Simple Model	Advanced Model
Churners caught	84 out of 100	**97 out of 100**
False alarms	54 out of 100 flagged	**Only 8 out of 100**
Campaign efficiency	Target 40% of base to catch all churners	**Target just 20%**



The advanced model identifies 97% of all future churners while keeping false alarms near zero — meaning the marketing team can run highly targeted campaigns without wasting budget on loyal customers.

Recommended 3-Tier Retention Campaign

Risk Tier	Customers	Action	Channel	Goal
High Risk	957 (98.4% churn)	Personal outreach + 15-20% discount + free shipping	Customer Service Call + VIP Email	Immediate Retention
Medium Risk	15 (6.7% churn)	Targeted coupon + product recommendations	Push Notification + Email	Re-engagement
Low Risk	4 649 (0.1% churn)	No intervention	---	Passive Monitoring



Strategic Recommendation:

Use the XGBoost model (AUC = 0.976 ± 0.013) to score all customers weekly.
Focus 80% of retention budget on the 957 High Risk customers — targeting just 17% of the base captures ~99% of all churners.